

From Recurrence to Attention

DATA 443/643 · Week 3, note 3.1

After Stanford CS224N W26 L5: the bottleneck, then a direct look at the source.

Lecture plan

- 1 The seq2seq bottleneck
- 2 Bahdanau rereads; Vaswani drops the loop
- 3 Path length 1 is not cost 1
- 4 Why a GPU likes attention

Key goal: leave with RNN path T , attention path 1, attention cost T squared.

First time: what / why / formula

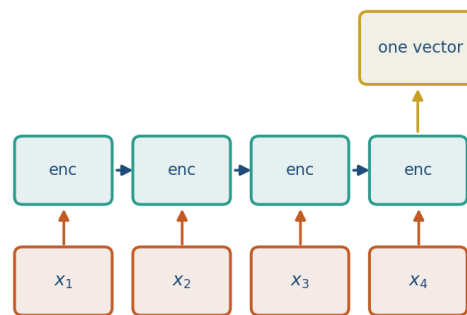
- What / why. Instead of squeezing a sentence into one RNN hidden state, each output position looks at all encoder states with a weighted sum. Seq2seq RNNs have a bottleneck: one vector for the whole source. Long sources forget the beginning.
- Architecture. Encoder states $h_1, \dots, h_T$. Decoder query q_t . Weights over source, then mix. Self-attention (next note) uses the same sequence for query and keys.
- Formula. Path length 1 at cost $O(T^2d)$. Bahdanau: $e_{ti} = v^\top \tanh(W_q q_t + W_h h_i)$.
- Tradeoffs. + No sequential bottleneck, interpretable weights. – Quadratic cost in T ; you still need the formula next hour.

How. Score, softmax, weighted sum. Cost is T^2 scores, not path length T .

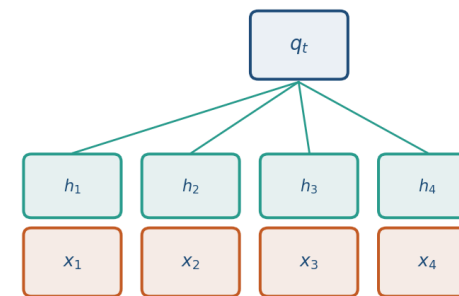
One vector is not a memory

- An encoder–decoder RNN packs the source into h_T .
- Long source, same-size bottle.
- Attention is a direct look at the encoder states.

Seq2seq bottleneck: the source is one vector



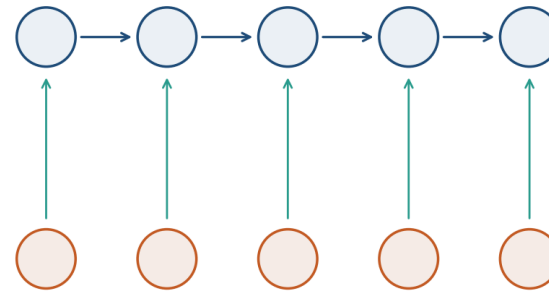
Attention: the decoder rereads the source



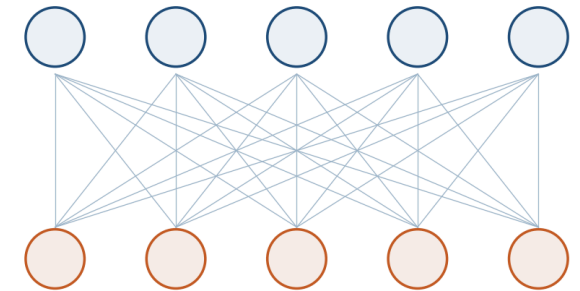
A chain versus all-pairs

- RNN: token 1 reaches token 40 through 39 overwrites.
- Attention: one score in a T by T map.
- Week 2's Jacobian product is no longer the story.

RNN: each state sees the past through a chain



Attention: every token can look at every token



Two different inventions

Bahdanau 2015

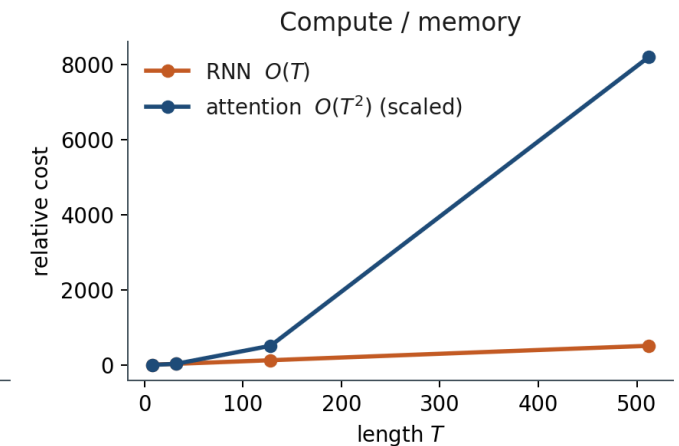
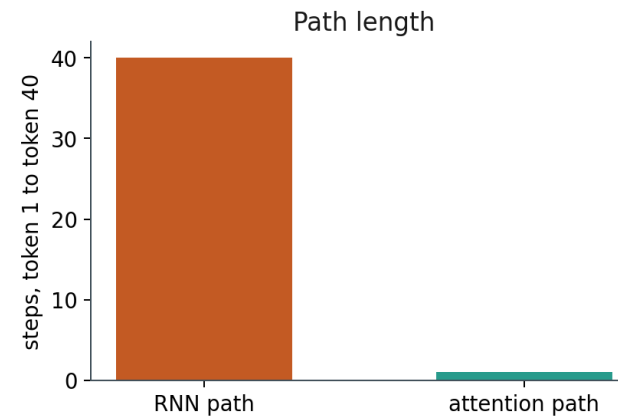
Attention on an RNN. Queries from the decoder; keys from the encoder.

Vaswani 2017

Attention instead of an RNN. Self-attention: Q, K, V from the same sequence.

Path length 1, cost T squared

- The short path is why we left recurrence.
- The quadratic bill is why Week 7 exists.
- Full attention is this course's default.



A 4k-token PDF

$$T = 4096, \quad T^2 \approx 1.68 \times 10^7$$

Scores per head per layer, before batching.

Four bytes each is about 67 MB for one map.

FlashAttention is an implementation, not a new model.

What to take from this lecture

Reread

Do not pack the source into one vector.

Path 1

Token 1 can touch token T in one hop.

Pay T squared

Lab 3 is $T=4$. A project PDF is not.